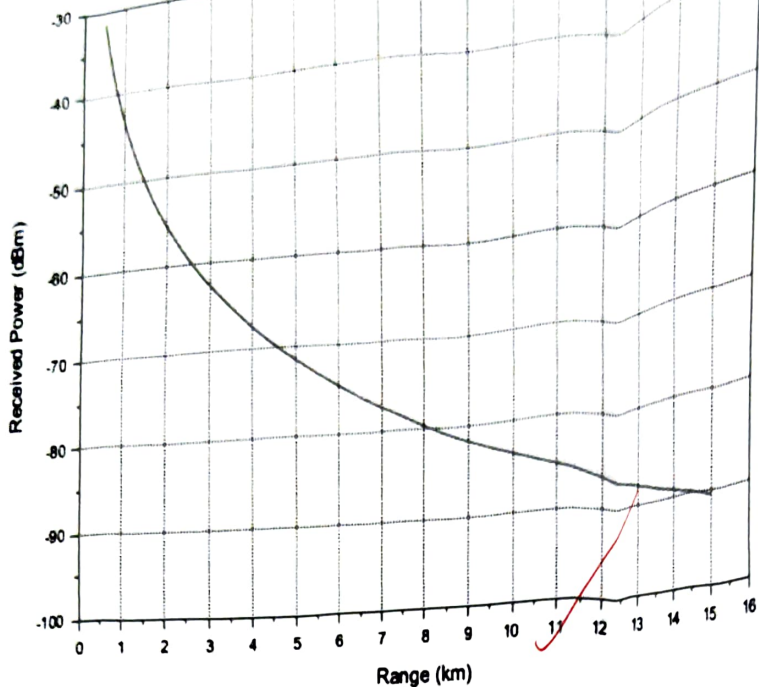
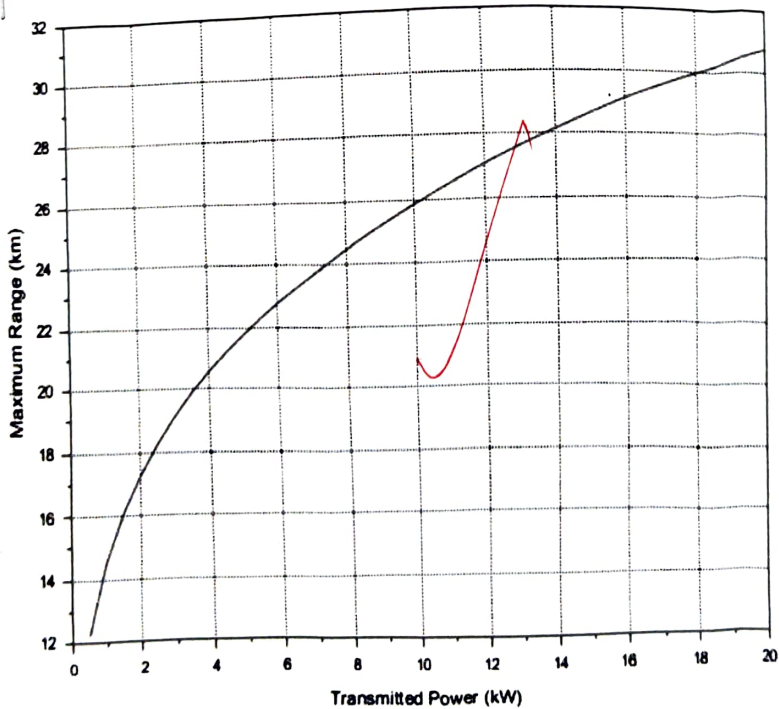


RADAR Received Power vs Target Range



Maximum RADAR Range vs Transmitted Power



5/9/26

EXP 8:-

AIM:-

To generate the plots for RADAR system in Sci-lab.

EQUIPMENT REQUIRED:-

Sci-lab

CODE:-

Range RADAR

clc;

clear;

$P_t = 1000;$

$G_{dB} = 30;$

$G = 10^{(G_{dB}/10)};$

$f = 10e9;$

$c = 3e8;$

$\lambda = c/f;$

$\sigma = 10;$

$L_{dB} = 3;$

$L = 10^{(L_{dB}/10)};$

$S_{min} = 1e-13;$

Watts

$$R_{max} = ((P_t * G^2 * \lambda^2 * \sigma) / ((4 * \%pi)^3 * S_{min} * L))^{(1/4)};$$

disp("RADAR RANGE EQUATION");

disp("-----");

disp("Transmitted Power (W):");

disp(P_t);

Parameter
for P_t variation

$$P_t = 6$$

$$G_t = 35$$

$$P_m = 10^{-12}$$

$$\lambda = 0.03$$

$$\sigma = 1$$

1)

for G_t variation

$$P_t = 2$$

$$G_t = 10$$

$$P_m = 10^{-13}, \lambda = 0.03$$

$$\sigma = 1$$

2)

for P_m variation

$$P_t = 2$$

$$G_t = 35$$

$$P_m = 6 \times 10^{-13}, \lambda = 0.03, \sigma = 1$$

3)

R_{max} (THEORETICAL)

$$R_{max} = \left(\frac{P_t G_t^2 \sigma}{(4\pi)^3 P_m} \right)^{1/4} = \frac{4 \times (35)^2 \times (10^{-03})^2}{(4\pi)^3 \times 10^{-15}} = 387.9 \text{ m}$$

R_{max} (PRACTICAL)

$$R_{max} = 390 \text{ m}$$

$$R_{max} = \left(\frac{(10)^2 (0.03)(12)}{(4\pi)^3 \times 10^{-13}} \right)^{1/4}$$

$$= 172.6 \text{ m}$$

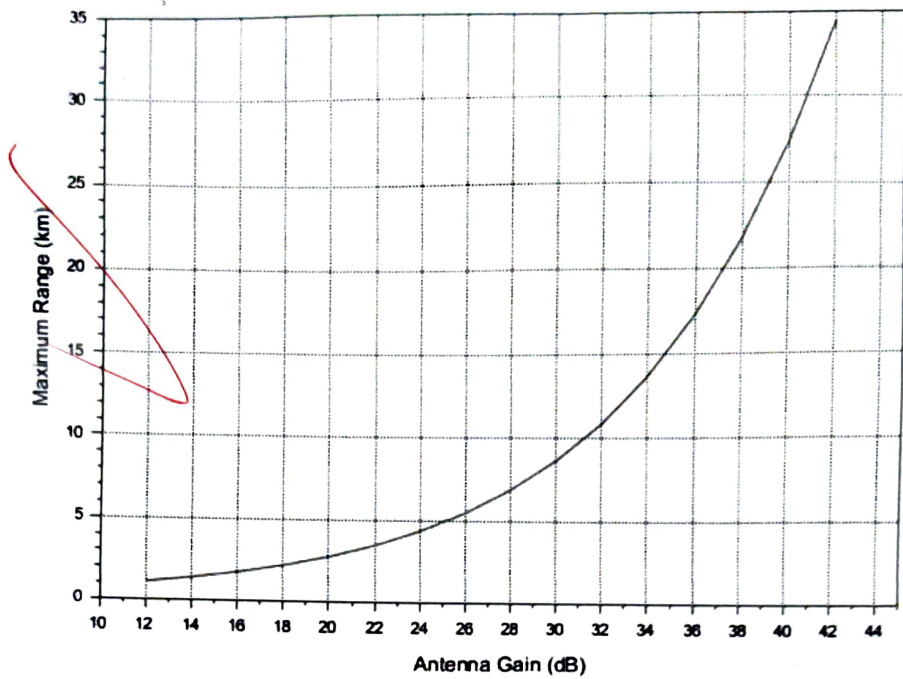
$$R_{max} = 175 \text{ m}$$

$$R_{max} = \left(\frac{(2(36)^2 (0.003))^{1/4}}{(4\pi)^3 (0^{-13})} \right)^{1/4}$$

$$= 72.6 \text{ m}$$

$$R_{max} = 75 \text{ m}$$

Maximum RADAR Range vs Antenna Gain



Scilab code & Implementat ⁿ (20)	Waveform and tabulation (20)	Calculation and result (20)	Viva- voce (20)	Result & observat ⁿ (20)
20	20	20	16	20 96

RESULT:-

The RADAR has been generated and plot has been achieved.

20/11/20